

Drive for better vision



EVK and PC Tool User Guide for HX6538 ISM028-03M V1.2

Himax Technologies, Inc
奇景光電股份有限公司

Agenda

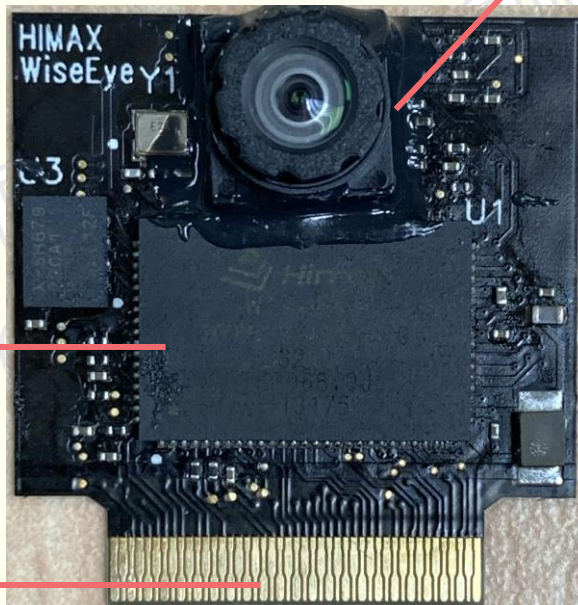
- **Board Introduction**
 - ❖ ISM028-03M Introduction
 - ❖ Demo Board 2(DB2) Introduction
- **PC Tool Usage**
 - ❖ PC Environment Setting
 - ❖ Flash Programming Procedures
 - ❖ Image Reception and Result Detection

Board Introduction

HX6538 ISM028-03M

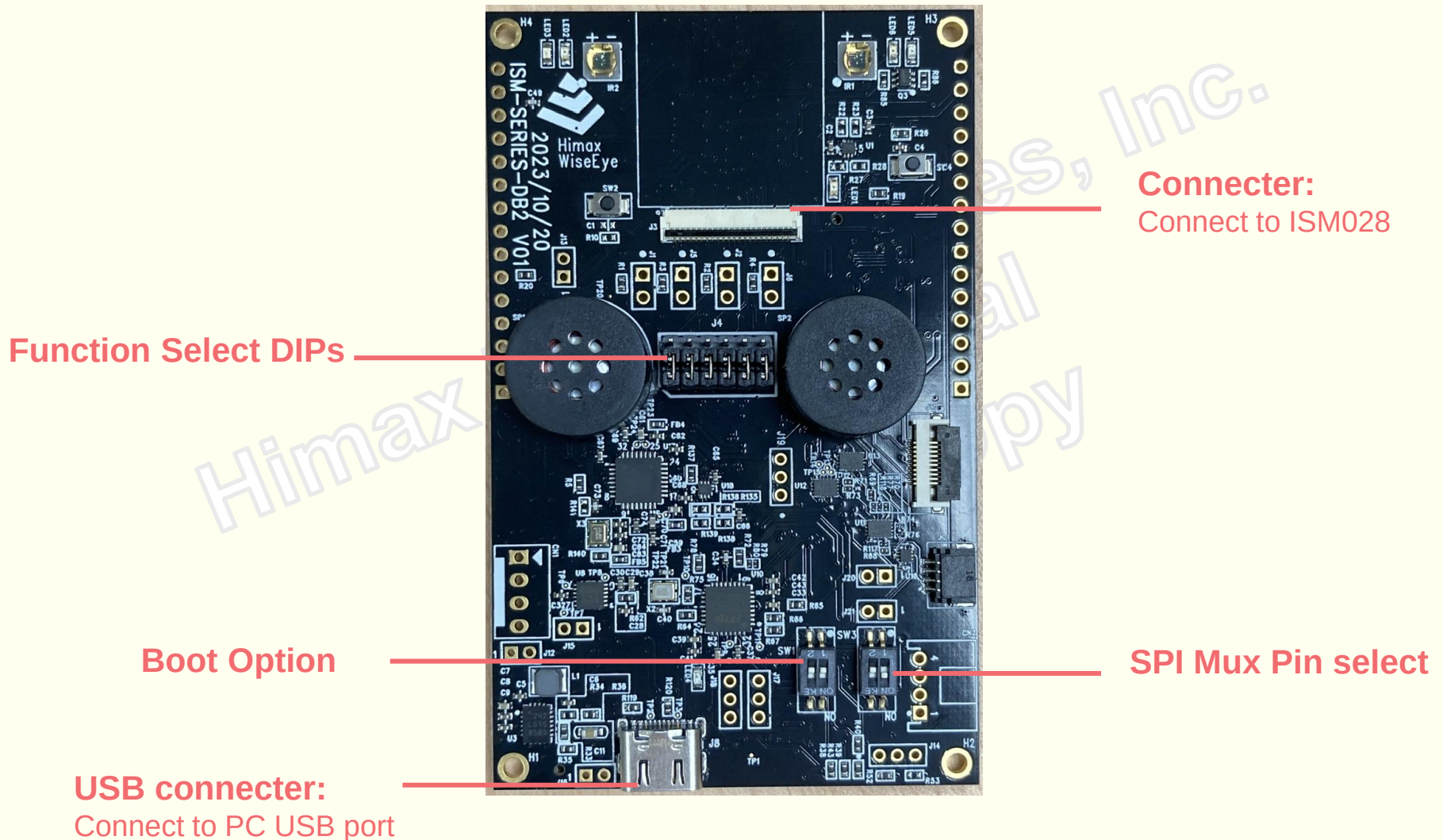
HX6538

Camera



Connector:
Connect to DB2

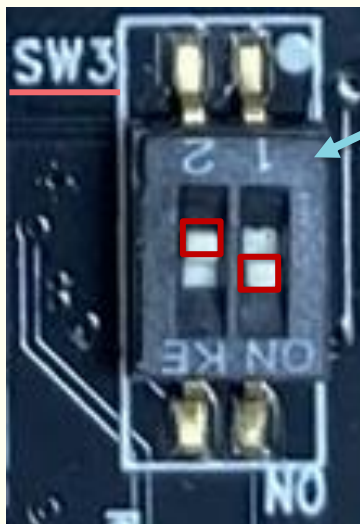
Demo Board 2 (FT4222H)



DB2 – SPI Mux PIN Select (SW3)

- **SPI Mux Pin select**

- ❖ [2, 1] = [OFF, **ON**]: FT4222H SPI Slave mode for image transfer
- ❖ [2, 1] = [**ON**, OFF]: FT4222H SPI Master mode for image transfer



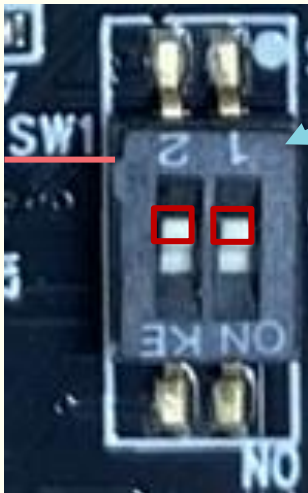
SPI Mux Pin select

SWITCH	MODE
 2 1	EVB: SPI Master (SDK Default) FT4222H: SPI Slave
 2 1	EVB: SPI Slave FT4222H: SPI Master

DB2 – Boot Option (SW1)

- **Boot Option Mode**

- ❖ [2, 1] = [OFF, OFF]: Normal Mode (Firmware Running)
- ❖ [2, 1] = [OFF, **ON**]: Flash Mode (For burning Firmware Image)

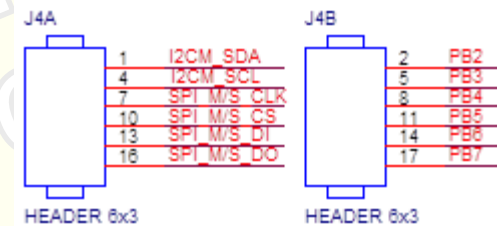


Boot Option select

SWITCH	MODE
 2 1	Normal Mode
 2 1	Flash Mode

DB2 - Function Select DIPs (J4)

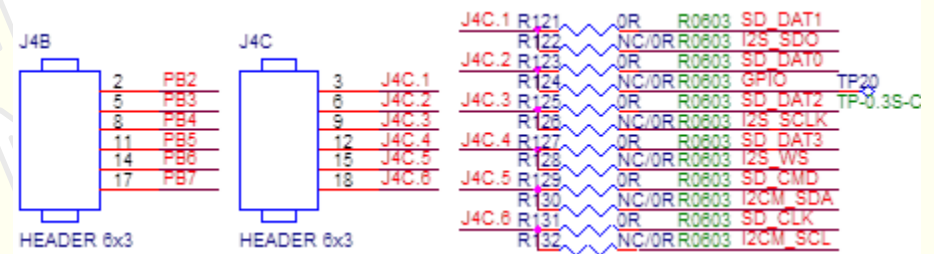
- J4B <-> J4A
 - ❖ PB2 <-> I2CM SDA
 - ❖ PB3 <-> I2CM SCL
 - ❖ PB4 <-> SPI M/S CLK
 - ❖ PB5 <-> SPI M/S CS
 - ❖ PB6 <-> SPI M/S DI
 - ❖ PB7 <-> SPI M/S DO



DB2 - Function Select DIPs (J4)

- J4B <-> J4C

- ❖ PB2 <-> SD_DAT1
- ❖ PB3 <-> SD_DAT0
- ❖ PB4 <-> SD_DAT2
- ❖ PB5 <-> SD_DAT3
- ❖ PB6 <-> SD_CMD
- ❖ PB7 <-> SD_CLK



PC Tool Usage

(*WE2_DEMO_TOOL*)

PC Tool Usage (WE2_DEMO_TOOL)

- **PC Environment Setting**

1. **Install FT4222H driver**

(path: SDK_root/_Documents/driver.7z)

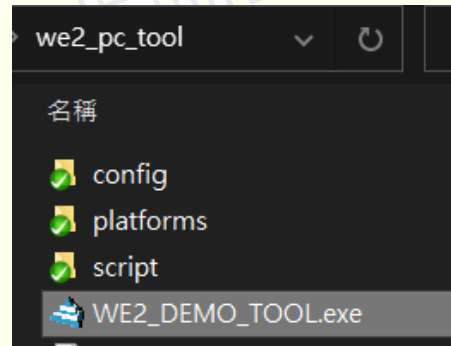
2. **Install VC++ Distribution**

(Download VC_redist.x64.exe by url:

<https://learn.microsoft.com/en-US/cpp/windows/latest-supported-vc-redist?view=msvc-170>)

- **Execute WE2_DEMO_TOOL.exe**

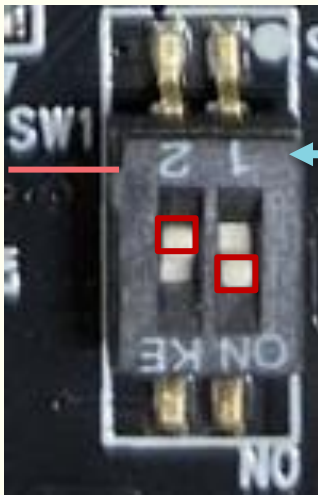
(path: SDK_root/we2_pc_tool/WE2_DEMO_TOOL.exe)



Flash Programming Procedures

Flash Programming Setting

- Set the switches on DB2 for **flash mode**.
 - ❖ [2, 1] = [OFF, ON]: Flash Mode (For burning Firmware Image)



SWITCH	MODE
 2 1	Flash Mode

EVB Flash Programming Procedures

1. Launch the PC Tool “**WE2_DEMO_TOOL**”
(path: SDK_root/we2_demo_tool/WE2_DEMO_TOOL.exe)

2. Following the procedures to flash firmware:

- ① Switch to “**BURN FLASH**” page.
- ② Click the “**Select File**” to select flash image file.
(path: SDK_root/we2_ota_tool/img/output.img)
- ③ Change Flash Type “**MXIC**”.
- ④ Click the “**Start**” to start burning image.
- ⑤ Wait the progress until 100%.

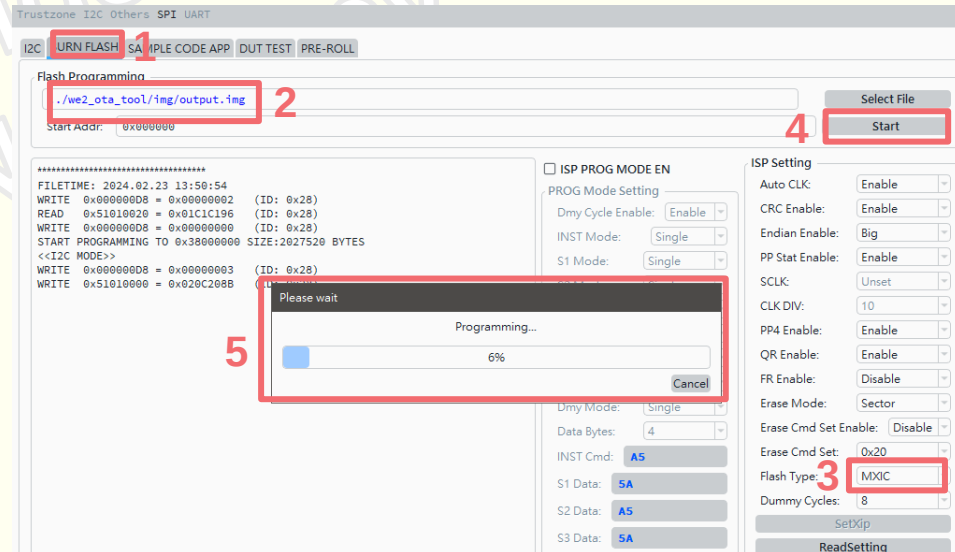
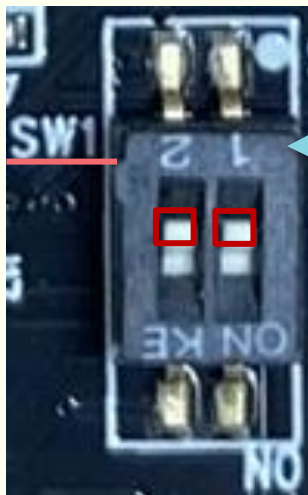


Image Reception and Result Detection

Normal Mode Setting

- Set the switches on DB2 for **normal mode**.



SWITCH	MODE
 2 1	Normal Mode

Image Reception and Result Detection

● Receive Image + Metadata

- ① Switch to “**SAMPLE CODE APP**” page.
- ② Choose “**SPI SLAVE**”.
- ③ Enable “**Enable Metadata**”.
- ④ Click “**START**” to start receiving the data from EVK.
- ⑤ PC Tool will show the image.
(If any person was detected, PC tool will mark the detected person with bounding box.)

